

Fetal Health Classification from Cardiotocogram Data

Reimplementation of "Multiclass Classification of Fetal Health using Cardiotocogram Data" (Ereyi & Huo, Stanford CS229, 2021)

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Problem and dataset

Problem

Classify each CTG record as Normal, Suspect or Pathological from numeric features. Manual reading is slow and subjective.

Data

UCI Cardiotocography: 2,126 records, 21 features, labelled by expert obstetricians. 2,113 used after preprocessing, reduced to 10 features(k-best feature technique that internally uses ANOVA) and standardized by removing mean and scaling to unit variance.

Class	Train	Test
Normal	1,149	497
Suspect	202	90
Pathological	128	47
Total	1,479	634

70/30 train-test split. About 78% of records are Normal.

Approach and implementation

Split data 70/30

**Select 10 features,
standardise**

Train 10 classifiers

**Compare with the
paper**

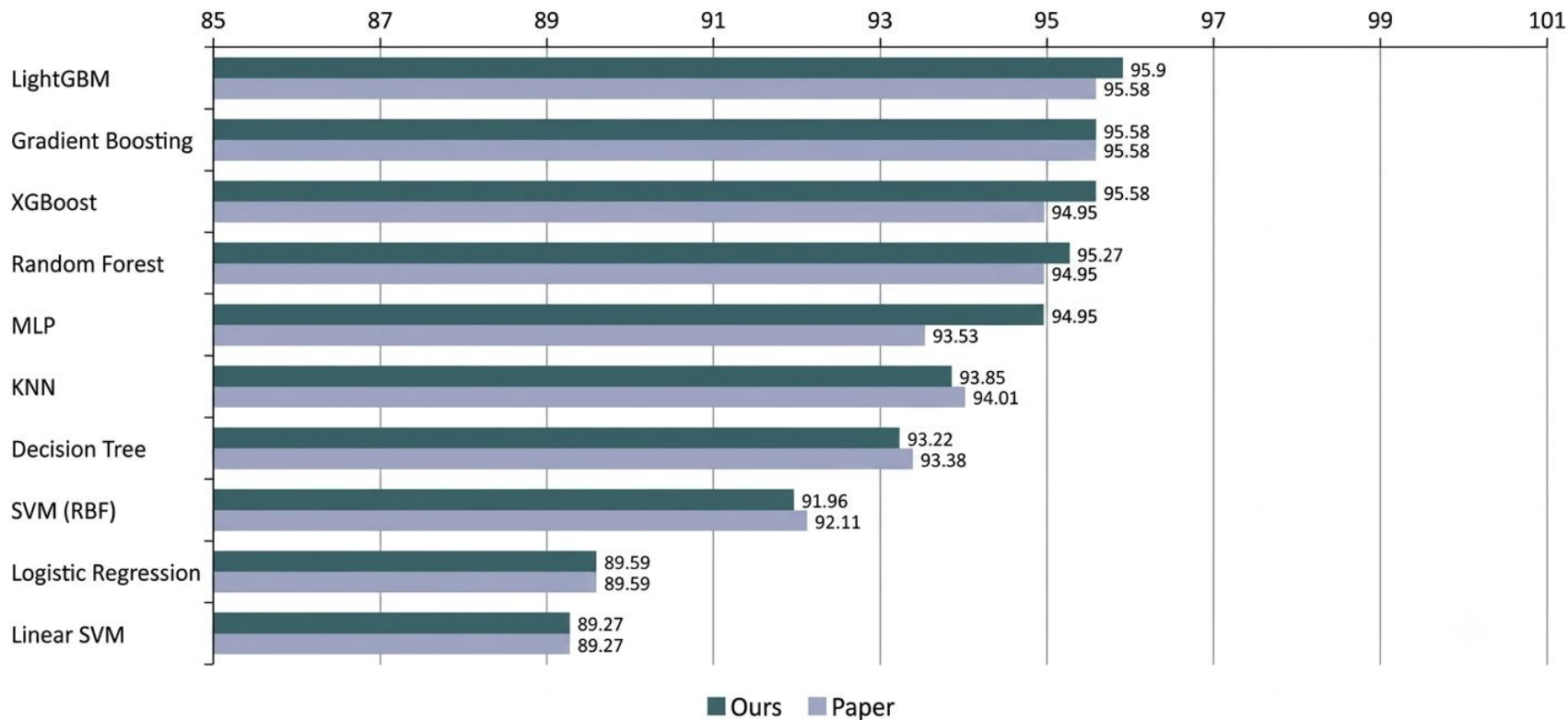
Notebook 03: MLP, Gradient Boosting, XGBoost, LightGBM, Linear SVM

Notebook 04: Logistic Regression, KNN, Decision Tree, Random Forest, SVM

Tools: Python, scikit-learn, XGBoost, LightGBM, Google Colab

Metrics: accuracy, per-class precision, recall and F1, confusion matrices

Accuracy: ours vs the paper



Best model: LightGBM

95.90%

test accuracy, 608 of 634 correct (paper:
95.58%)

Class	Precision	Recall	F1
Normal	0.967	0.990	0.978
Suspect	0.947	0.789	0.861
Pathological	0.900	0.957	0.928

Suspect is the hardest class: it is small and overlaps with Normal.

Conclusions

- Boosted trees and Random Forest were best (95.3 to 95.9%). Linear models stayed near 89%.
- Nine of ten models are within 0.65 points of the paper. The MLP differs most (+1.42), as the paper does not give its size or seed.
- One test sample is 0.16 points, so small gaps are noise.
- Limitation: a single split, no cross-validation.